

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Step 1: Problem Discovery

Observation

Students often struggle silently when learning. They find it difficult to:

- Identify their own emotional state during study sessions
- Communicate their struggles effectively to educators
- Receive timely, personalized support tailored to their mindset

Who is Affected?

Stakeholder	Pain Point
Students	Cannot articulate study problems; receive generic help that doesn't match their
Teachers	Unable to monitor emotional well-being of individual students at scale
EdTech Platforms	Lack personalization based on learner's current emotional and cognitive state

Step 2: Problem Statement (HMW Format)

"How might we help students receive emotionally-aware, personalized learning support by automatically detecting their emotional state from the way they describe their study problems?"

Why #	Question	Answer
Why 1	Why do students struggle silently?	They don't know how to express their emotional state to educators
Why 2	Why can't they express it?	
Why 3	Why don't tools detect emotions?	Traditional learning tools don't prompt or detect emotional context
Why 4	Why is affect recognition missing?	
Why 5	Why is it complex?	Most EdTech focuses on content delivery, not learner affect recognition
		Integrating NLP emotion detection into learning platforms is technically complex
		There is no accessible, off-the-shelf system that combines emotion detection wit

Step 4: Problem Statement (Formal)

> Problem: Students in digital learning environments lack access to an intelligent system that can detect their emotional state from natural language descriptions of their study problems and automatically generate contextually appropriate, personalized learning guidance.

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> Impact: This results in reduced learning outcomes, increased dropout rates, and student burnout - particularly for self-paced learners with no immediate access to a teacher.

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> Proposed Direction: Build an AI system that classifies 5 student emotions (Bored, Confident, Confused, Curious,

Step 5: Acceptance Criteria

Criterion	Target
Emotion classification accuracy	80% on validation set
Response generation latency	< 5 seconds per query
Emotions supported	5 (Bored, Confident, Confused, Curious, Frustrated)
Deployment platform	Accessible via public URL (Streamlit Cloud)
User interface	Intuitive, no technical knowledge required